PEAS Framework and Task Environment Analysis

System:PersonalizedNewsFeedRecommender(Twitter/X · TikTokAlgorithm)

Task 1: PEAS Specification

Performance Measure

Four metricsdefinerecommender performance:

- Engagement rate (meaningful interactions only, not raw clicks)
- Session satisfaction (feedback + return frequency)
- Content diversity (topic/source/perspective spread)
- Misinformation suppression (down-ranking flagged content before spread)

Environment

Theagent operates in a global digital ecosystem. Locally, it adapts to device context such as time, network quality, screen size, and user scrolling behavior. It also processes a massive, multilingual stream of content published at scale across many sources, while leveraging dynamic social signals.

External events like breaking news or viral activity introduce sudden shifts, and platform constraints including advertising rules, regulation, and moderation further limit what can be surfaced safely and legally.

Actuators

Rankedcontent feed renderer: Determines position, preview size, and autoplay behaviour placement alone shifts engagement by orders of magnitude.

Push notification dispatcher: Delivers breaking content proactively; must be metered to avoid trust erosion.

A/B test allocation engine: Routes user cohorts into controlled experiments for autonomous comparison of ranking variants.

Content diversity injectors: Insert serendipitous items outside dominant interest clusters, counteracting filter bubble collapse.

Feedback signal collector: Records explicit reactions (likes, dislikes, hides, reports) and implicit ones (dwell time, scroll-past velocity).

Sensors

Interaction telemetry streams: Every tap, scroll pause, share, and skip at millisecond resolution revealed preference far more reliably than stated preference.

NLP embeddings: Encode content into high-dimensional semantic space, enabling similarity matching beyond keyword overlap.

Real-time trending signal detector: Monitors engagement velocity, identifying breaking topics before they peak.

Social graph propagation trackers: Observe content movement through follow networks, distinguishing organic virality from coordinated amplification.

User context sensors: Device type, geolocation (with consent), local time, and ambient language adapt recommendations to situational relevance.

Source credibility scorer: Ingests third-party fact-check verdicts and historical accuracy records to modulate promotion of new outlet content.

Task 2: Environment Classification

Dimension	Classification	Justification
1. Observability	**Partially Observable**	The agent cannot directly access a user's cognitive state, mood, or offline context only their behavioural trace. A scroll-past may mean disinterest or prior reading elsewhere; the system cannot distinguish these.
2. Agent Count	**Multi-agent**	Millions of creators, advertisers, bot networks, and recommendation systems act simultaneously. A coordinated campaign by one actor can distort the signal environment for every other agent.
3. Determinism	**Stochastic**	User response is inherently probabilistic identical content shown at different times produces different outcomes. External events add additional uncontrollable variance.
4. Episode Structure	**Sequential**	Each recommendation conditions the next: surfacing a divisive article shifts emotional state and changes subsequent content appeal. Treating sessions as independent episodes would produce incoherent feeds.
5. Temporal Stability	**Dynamic**	The content pool refreshes continuously, trending topics peak and decay in hours, and user preferences drift over weeks. Any static model becomes stale rapidly.
6. State Space	**Continuous**	User preference is a dense embedding vector, article relevance is real-valued, and engagement probability is a continuous output. Discrete approximations exist only at the UI layer.

Task 3: Critical Analysis

Most Challenging Environmental Property

The key challenge is dynamics, which is harder than partial observability because it makes past data quickly outdated. Preferences shift so fast that yesterday's model can become misleading. The system must balance two opposing needs: slow learning from long-term signals for stable preferences, and fast adaptation to real-time trends. Stability improves reliability but lags behind change, while real-time responsiveness is fast but noisy and vulnerable to manipulation. The core problem is safely combining both in production without manual control.

Utility Function and Trade-off

$$U = \alpha \cdot \textbf{(Engagement Rate)} - \beta \cdot \textbf{(Filter Bubble Index)} - \gamma \cdot \textbf{(Misinformation Risk)}$$

In a well-calibrated system, the three weights balance engagement, ideological diversity, and misinformation control. Small engagement losses are acceptable if they reduce echo chambers and harmful exposure. Increasing α too much quickly pushes the model toward outrage- and tribe-driven content, reducing diversity and resurfacing flagged sources in high visibility.

Users then become trapped in narrow content diets, reducing exposure to counter-narratives and degrading future training data quality. Overweighting any single factor leads to long-term systemic harms that are not visible in short-term metrics.

Conclusion

The personalized news feed recommender operates at the confluence of the hardest environment properties: partial observability of user intent, a multi-agent ecosystem where adversarial actors compete for attention, stochastic user response, sequential session dynamics, and a content world that changes faster than any static model can track. The PEAS analysis makes clear that each design decision from sensor architecture to utility function weights must be made with all six dimensions in view simultaneously. Systems that optimise engagement in isolation produce engagement; systems that optimise for the full PEAS picture produce something closer to a genuinely useful information service.

Task 4: Structured Data Design - Structured Representation

The following JSON object encodes the PEAS specification (Task 1) and environment classification (Task 2) in a single machine-readable structure, satisfying all assignment requirements: top-level keys ***system_name***, ***peas***, ***environment_classification***, and ***utility_function***; at least four performance measures stored as an array; and each environment dimension represented as a nested object with ***choice*** and ***justification*** fields.

```
{
"system_name": "Personalized News Feed Recommender",
"peas": {
"performance_measures": [
"engagement_rate_meaningful_interactions_only",
"session_satisfaction_feedback_and_return_frequency",
"content_diversity_topic_source_perspective_spread",
"misinformation_suppression_flagged_content_downranked"
],
"environment": "Global digital ecosystem with device-level context (time, network,
screen size, scrolling behaviour), multilingual content streams, dynamic social
signals, breaking-news events, and platform advertising/moderation constraints",
"actuators": [
"ranked_content_feed_renderer",
"push_notification_dispatcher",
"ab_test_allocation_engine",
"content_diversity_injector",
"feedback_signal_collector"
],
"sensors": [
"interaction_telemetry_streams_millisecond_resolution",
"nlp_embedding_encoder_high_dimensional_semantic_space",
"real_time_trending_signal_detector",
"social_graph_propagation_tracker",
"user_context_sensors_device_geo_time_language",
"source_credibility_scorer_factcheck_verdicts"
]
},
"environment_classification": {
"observability": {
"choice": "Partially Observable",
"justification": "The agent cannot access a user's cognitive state or offline
context; a scroll-past may signal disinterest or prior reading — indistinguishable
from behavioural trace alone."
},
"agents": {
"choice": "Multi-agent",
"justification": "Millions of creators, advertisers, and bot networks act
```

```
simultaneously; a coordinated campaign by one actor distorts the signal
environment for every other agent."
},
"determinism": {
"choice": "Stochastic",
"justification": "Identical content shown at different times produces different
outcomes; external events add uncontrollable variance to user response."
},
"episode_structure": {
"choice": "Sequential",
"justification": "Each recommendation conditions the next; surfacing a divisive
article shifts emotional state and changes subsequent content appeal."
},
"temporal_stability": {
"choice": "Dynamic",
"justification": "The content pool refreshes continuously, trending topics peak
and decay in hours, and user preferences drift over weeks."
},
"state_space": {
"choice": "Continuous",
"justification": "User preference is a dense embedding vector; article relevance
and engagement probability are real-valued continuous outputs."
}
},
"utility_function": "U = alpha*(Engagement_Rate) - beta*(Filter_Bubble_Index)
- gamma*(Misinformation_Risk)"
}
```

The utility function encodes the core design trade-off: maximising engagement while penalising filter-bubble formation and misinformation exposure. If the weight α (engagement) were doubled, the agent would increasingly favour outrage- and tribe-driven content, surfacing flagged sources and collapsing content diversity illustrating why balanced weight calibration is essential for a trustworthy recommender system.